STUDENT MARKS ANALYSIS USING PYTHON

1. Introduction
Student Marks Analysis is a Python-based project designed to analyze student scores and summarize performance through simple statistical calculations. It demonstrates how programming can be used to process numerical data and present meaningful results.

2. Objectives
- Calculate total and average marks.
- Identify the highest and lowest marks.
- Analyze individual student performance.
- Categorize student scores.
- Represent results visually.

3. Tools and Technologies
Python, Google Colab, Jupyter Notebook, Pandas, and Matplotlib.

4. Dataset Description
The project uses five sample student records. Each student has a mark between 0 and 100. The sample dataset is intended only for demonstration.

5. Methodology
Student names and marks are stored in Python lists. The program validates the inputs and uses built-in functions to calculate total marks, average marks, highest marks, and lowest marks. It then categorizes performance, creates a results table, generates a bar chart, and exports the results to CSV.

6. Results
- Total students: 5
- Total marks: 436
- Average marks: 87.20
- Highest marks: 95
- Lowest marks: 76
- Marks range: 19

The sample data identifies Vasanth as the highest scorer and Arjun as the lowest scorer.

7. Conclusion
The project successfully demonstrates basic Python programming and statistical analysis. It shows how a simple program can summarize data, compare individual scores, and visualize student performance.

8. Future Enhancements
The project could be extended to read student records from Excel or CSV files, support multiple subjects, calculate grades using configurable rules, and generate more detailed reports.

9. Repository

GitHub repository: https://github.com/venky1318/student-marks-analysis